# 21M.370 Digital Instrument Design
# Lab assignment 3

**Goals:**

1. Assemble and test the electronic components for our first class instrument
2. Use the components to control parameters in your python code from lab 2

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**Assignment description:**

Our goal for this week is to assemble the electronic components needed for our first instrument, Nobby, and to create a simple test mapping. At the conclusion of this lab, you will have:

1. 4 pots and 4 buttons mounted in an enclosure and connected to the 370 breakout like:
   1. Pots connected to inputs 0, 1, 2, & 3
   2. Buttons connected to inputs 6, 7, 8, & 9
2. uPython installed on your ESP32
3. a script written for your ESP32 to test your sensors
4. a simple script written to send midi from the esp32 to the python script on your desktop
5. A modified version of your lab 2 python code to handle midi input
6. You can use any PD patch you want.

Several video tutorials are available to help. They walk through all both troubleshooting / foundational explanations as well as explain how all of the software works.

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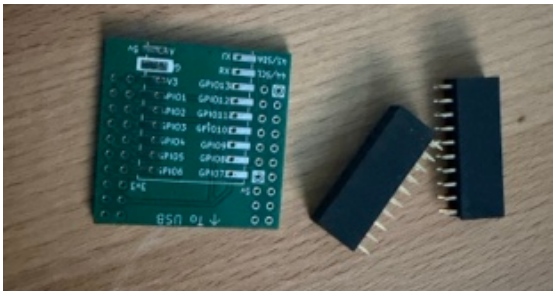
**Assembling all of your components:**

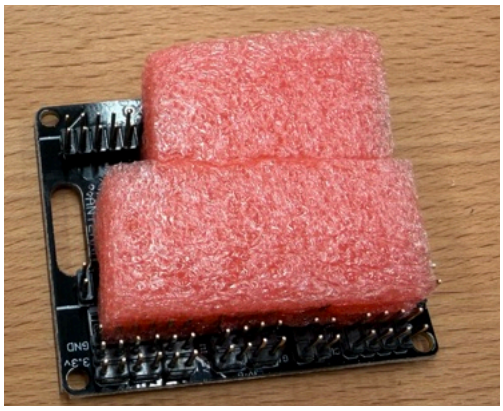
For this lab you will need:

1. A cardboard box

2. An ESP32s3 Zero

3. An m370 breakout PCB and socket headers to fit into it
4. An ESP32 adaptor PCB to fit the ESP32 on to our breakout board

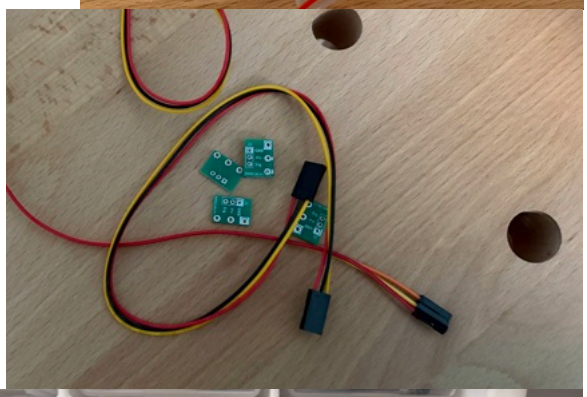




5. 4 buttons

6. 4 potentiometers and knobs
    1. we can get more knobs if we need them
    2. the pots are in the cardboard box as well as the plastic container below them.
    3. Note: be sure to use only B-style (linear) pots! That should be all of them but it doesn't hurt to check.

7. 4 potentiometer adaptor PCBs
8. socket-to-socket jumper cables



9. pin headers
   - they will be in the white plastic parts organizers near the soldering station.

10. usb-C cable
    1. we don't have a ton right now - hopefully you have one you can use?

All of these components will be on the shelf in W18-2307, the music tech makerspace.

**Soldering the ESP32 and sensors:**

The following video tutorials cover basic soldering and electronics for our first instrument:

Main class youtube playlist
Soldering the ESP32
Assembling potentiometers
Assembling buttons
Plugging into the breakout board
Basic electronics

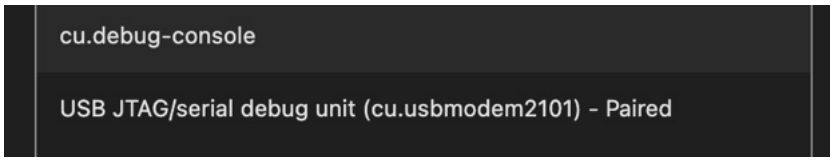
You can solder anywhere, but the music tech makerspace is a great place to do it.

**Install circuitpython software on the ESP32**

New for this semester, we will be installing circuitpython on our ESP32 and programming it in python! The first step we will only need to do once - and you will probably need to do it in google chrome.

1. Go to https://circuitpython.org/board/waveshare_esp32_s3_zero/
2. Click “Open Installer” (if nothing happens try it in chrome)
3. Click “Install CircuitPython 10.1.3 Bin Only“
4. Click Next

5. Connect to your board. Click the “Connect” button, and a popup should appear that looks something like: USB JTAG serial debug / usbmodem.
    1. You might need to explore the different options provided for you.



6. Click connect
7. Click continue to erase flash (optional) and install the bin file
8. Hopefully this completes successfully
9. Unplug and replug the esp32.
10. You should see the LED light up and start to blink.

Done with installing the firmware! Now we can program our ESP32 in python.

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